

Mathematics for Machine Learning

<https://mml-book.github.io/book/mml-book.pdf>

1 Linear Algebra

- Polynomials form a vector space.
- Elements of $\mathbb{R}^n$ (tuples of n real numbers) are vectors. For example,

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3.$$

- Adding two vectors $a, b \in \mathbb{R}^n$ component-wise results in another vector:
 $a + b = c \in \mathbb{R}^n$.
- Scaling: $a \in \mathbb{R}^n, \lambda \in \mathbb{R} \implies \lambda a \in \mathbb{R}^n$.